Before Worrying About a Gravitational Ghost, Ask Whether It Is Really There

A general-audience guide to ghosts, gauge reduction, and conformal gravity

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Einstein changed our picture of gravity completely. Gravity is not an invisible force pulling objects through space. It is the bending and stretching of spacetime itself.

That idea works extraordinarily well, from falling apples to black holes and the expansion of the universe. But it also creates one of physics' hardest problems: how do we combine a dynamical spacetime with quantum physics?

One tempting approach is to modify Einstein's equations by adding terms involving higher powers of spacetime curvature. These additions can improve the theory's behavior at extremely short distances, where quantum effects become important. Unfortunately, they also introduce extra solutions—often called gravitational "ghosts."

A ghost is not a supernatural object. It is a mathematical warning sign. Depending on the formulation, a ghost may appear as a mode with negative energy, negative probability, or the wrong sign in the inner product used to calculate probabilities.

For decades, this has encouraged a simple conclusion:

> Higher-derivative gravity contains ghosts, so it cannot be a sensible theory.

Our work suggests that this conclusion is too quick. Not because the ghost problem has disappeared, but because "the ghost problem" is actually several different problems that are often mixed together.

The most basic lesson is this:

> Before asking whether a gravitational mode has positive or negative norm, we must first determine whether it survives the complete gauge reduction at all.

Three different questions

A higher-derivative theory must pass at least three logically separate tests.

The first concerns its local equations. What solutions do the equations allow? What frequencies and particle-like modes appear? Can the free theory be given a positive notion of probability?

The second concerns gauge symmetry. Gravity contains many mathematical descriptions of the same physical situation. Once all relevant redundancies have been removed, which of the apparent modes remain as independent states or observables?

The third concerns interactions. Even if a free theory has a satisfactory description, does that description survive when the modes exchange energy and interact?

These questions are connected, but they are not interchangeable.

A mode can appear in the field equations and then disappear after gauge reduction. Conversely, a free theory can have an elegant positive description that breaks down when exact energy-conserving interactions are introduced.

Our eight-paper technical programme separates these issues rather than trying to settle them with one slogan.

A lesson from a mechanical toy model

The programme begins with the Pais–Uhlenbeck oscillator, a famous mechanical model with higher time derivatives. It is much simpler than gravity but contains the same basic ghost puzzle.

In one description, the oscillator appears to contain a negative-energy mode. In another, obtained through a non-standard change of variables and inner product, it behaves like two positive oscillators.

This does not mean that every ghost can be removed by clever notation. It means that the complex equations alone do not determine which quantities should be regarded as real, which operators are observable, or how probabilities should be calculated.

The positive description also has limits. When the two frequencies merge, the transformation used to construct it becomes singular. And when interactions permit exact conversion between different branches, the free positive structure can encounter genuine obstructions.

This led us to separate the geometry of the free theory from the stability of that geometry under interactions.

Gravity adds another ingredient: gauge symmetry.

Why gauge symmetry changes the question

Suppose you describe the same point on Earth using two different maps. The coordinates differ, but the place does not.

Gravity works similarly. A change of spacetime coordinates can alter the mathematical appearance of the gravitational field without changing the physical geometry. Conformal gravity has an additional symmetry: certain local changes of scale also leave the essential structure unchanged.

A raw list of solutions therefore overcounts the physical possibilities.

Physicists use a framework called BRST theory to keep track of this systematically. BRST introduces special bookkeeping variables—also, confusingly, called "ghosts"—for each gauge symmetry. These BRST ghosts are not the negative-energy particles discussed above. They are algebraic tools for separating genuine information from redundancy.

The calculation asks two questions:

- Is an object compatible with all the gauge constraints?
- Is it merely a gauge transformation of something else?

What survives both tests is called cohomology. In ordinary language, cohomology is the part of the theory that remains after every certified redundancy has been removed.

Why put conformal gravity on a cylinder?

Our new result concerns free pure-Weyl gravity, a particularly symmetric form of four-dimensional conformal gravity.

We study it on a mathematical spacetime called the conformal cylinder: time runs along an infinite line, while each spatial slice is a three-dimensional sphere.

This is not a claim that the real universe literally has this shape. The cylinder is a powerful lens for studying conformal symmetry.

Because the spatial sphere is compact, the gravitational field breaks into a discrete collection of modes instead of a continuous range of momenta. The cylinder also organizes all fifteen conformal transformations—translations, rotations, scale transformations and their relatives—into one geometric structure.

There is then an important physical choice.

One may keep some of these transformations as ordinary global symmetries with measurable charges. Alternatively, in a closed-universe description with no spatial boundary, one may treat all fifteen as residual gauge transformations and impose their charges as constraints.

Our theorem studies the second choice, but a later charge calculation makes the restriction sharper. On the unrestricted compact phase space, cylinder time translation carries a charge and is not gauge. The theorem applies on the full zero-charge, or Taub, sector and its selected derived quotient. In that sector the time-translation generator is a constraint.

If time translation is instead retained as a physical Hamiltonian—for example because one introduces a boundary, an external clock or an asymptotic region—the problem changes and our reduction does not apply unchanged.

What happens when we add a clock?

A natural objection is that any usable clock should immediately turn time translation into a physical symmetry. We tested that directly rather than assuming an answer.

Several simple clock constructions fail. A single standard scalar cannot provide the required exact cylinder background, and a two-field cancellation can make the total charge vanish only by introducing an unhealthy local degree of freedom.

The first healthy counterexample uses a Berger cylinder: the spatial three-sphere is smoothly squashed rather than perfectly round. Two standard-sign rotating scalar fields support this background and provide a monotone internal clock. The matter clock has genuine nonzero internal momentum. Nevertheless, at fixed couplings and to linear order, the full constraint equations force the variation of the total time-translation charge to vanish.

This proves a possibility, not a universal rule. A healthy clock does not *automatically* make total time translation physical. We still have to construct explicit relational observations—such as a clock-defined redshift—and determine whether the result survives general perturbations, interactions and quantization.

Closing the chain from fields to residual states

Earlier stages of the project had computed the final residual cohomology using an exact algebraic model. The remaining concern was whether this model really represented the complete classical field theory or merely resembled it.

That gap is now closed in the selected algebraic category.

The construction starts from the actual gravitational field, its coordinate and scale symmetries, and all the corresponding BV variables. BV theory is an extended accounting system that includes fields, gauge transformations, equations, identities and their dual partners in one complex.

We then:

1. derive the free BV complex directly from the pure-Weyl action;
2. separate and contract variables that contain no independent information;

3. preserve all fifteen residual conformal zero modes;
4. transfer those zero modes into one residual BFV constraint system on a spatial slice;
5. choose the positive-frequency half of the classical phase space as the state polarization;
6. compute the resulting residual BRST cohomology and its pairing.

The resulting state complex is no longer a hand-selected collection of curvature modes. It is derived from the metric field theory through an explicit, verified chain.

The causal bridge is now complete on the cylinder

An algebraic reduction is not enough by itself. A field theory must also say how disturbances propagate and whether its gauge constraints remain consistent under that propagation.

The second cylinder paper now supplies that bridge for the complete free classical theory. After the fields, gauge transformations, equations, identities, curvatures and auxiliary variables are organized into one large complex, we construct retarded and advanced chain homotopies on every row. In ordinary language, the full gauge system has consistent past- and future-directed response maps with causal support.

This required working with the entire complex. The isolated fourth-order metric equation cannot be converted into the preferred simple scalar-wave block within the tested complete families. Curvature and tractor variables provide the successful larger system. That system transports the causal Green pairing, the covariant current pairing and the residual pairing into one another, and it reproduces the same two curvature-square classes with the same positive matrix.

This is a causal theorem on the conformal cylinder. It is not yet a theorem on arbitrary curved spacetimes, at black-hole horizons or at the boundary of an asymptotically flat universe.

What survives?

The result is striking.

In the relevant centered sector, no individual one-particle conformal-graviton class survives the residual gauge reduction.

The original one-particle module is indefinite: it contains both positive- and negative-sign directions. But the calculation does not repair those signs. Instead, the entire one-particle sector disappears from this particular cohomology.

The first surviving objects are composites made from two Weyl curvatures. There are exactly two.

One combines two curvatures of one chirality—the gravitational analogue of handedness. The other combines two curvatures of the opposite chirality.

Their even and odd combinations correspond to two familiar geometric terms:

- the square of the Weyl curvature;
- the Pontryagin density, which records topological information.

The first changes the bulk gravitational equations and is therefore a genuine dynamical direction. The second is locally a boundary term. It does not create an independent bulk equation, although it may still matter when topology, boundaries or gravitational "theta" data are present.

The two classes have a normalized positive pairing. In matrix language, their Gram matrix is the two-dimensional identity matrix.

That statement needs careful interpretation.

It does not mean that we have found two positive-norm graviton particles. It means that two ghost-dressed curvature-square vertex classes have positive pairing in the resulting classical cohomology.

They are not one-particle gravitons.

They are best thought of as possible directions in the space of actions, interactions or observables—not as two particles flying through space.

What the result changes

The usual ghost discussion begins by looking at the signs of the local oscillator modes.

Our result shows why that can be premature. A locally visible mode is not automatically an independent physical state. Gauge reduction can change the object being studied before any question about probability is asked.

This does not prove that the conformal-gravity ghost problem has been solved. It sharpens the question.

Instead of asking only:

> Can the negative-sign conformal graviton be given a positive norm?

we should first ask:

> After specifying the boundary conditions and performing the full gauge reduction, what is the physical object on which a positivity question should be asked?

For the closed-cylinder reduction studied here, the answer is not a one-particle conformal-graviton space. It is a two-dimensional space of classical curvature-square deformation classes.

Why interactions remain a separate test

The other branch of our programme studies interactions.

There, the lesson is almost complementary. A free theory may possess a positive mathematical description, yet exact energy-conserving conversion processes can obstruct the continuation of that description once interactions are switched on.

Gauge reduction asks which objects survive the constraints. Interaction analysis asks whether the surviving structures remain dynamically consistent.

Neither question replaces the other.

The first clock-coupled nonlinear test has now passed through quadratic order: the complete binary interaction operation is cyclic, and the corresponding time-translation contraction can be constructed on the full Berger bookkeeping complex. That is encouraging, but it is not full nonlinear stability. Cubic order, physical resonant channels and the analytic estimates needed for the next causal construction remain open. One proposed analytic import is deliberately marked fail-closed until those estimates and its source-versus-solution maps are certified.

We have also verified that familiar Einstein–Maxwell gravitational and electromagnetic wave modes occur inside the larger Weyl–Maxwell linear solution space on a compact product background. Their induced pairing is nondegenerate, but it is not simply the Einstein pairing and some fixed-charge modes encounter genuine second-order obstructions. This is evidence that the larger theory contains ordinary gravitational physics, not yet proof that the Einstein sector is the complete or stable physical sector.

What has not been proved

The core result is classical and free. It began with finite combinations of definite cylinder-energy modes in a specified closed-universe polarization. That mode space has now also been completed to allow controlled infinite energy-mode sums, without changing the two surviving classes. The reduced field equations have a matching fixed-time wave description.

It does not yet provide:

- an extension of the covariant causal theorem beyond the conformal cylinder to generic globally hyperbolic backgrounds;
- a general distributional or Hadamard-state construction;
- a proof that the construction survives nonlinear gravitational evolution;
- a solution of the quantum master equation;
- cancellation of quantum Weyl or gravitational anomalies;
- a determination of what happens to the two classical classes under renormalization;
- a positive graviton Fock space;
- an asymptotically flat Bach phase space, black-hole boundary theorem, or particle and scattering-matrix unitarity.

It also does not explain dark matter or dark energy. Its subject is more foundational: the mathematical consistency and interpretation of candidate quantum theories of gravity.

The next question

The free classical chain is now complete in its stated cylinder domain:

> gravitational fields → BV gauge complex → residual BFV constraints → causal propagation and pairing → polarized state complex → two curvature-square cohomology classes.

The frontier has moved.

The next challenge is to find the boundary of that result: add relational clocks, physical interactions, general backgrounds, asymptotic charges and quantization without silently changing what is being called gauge.

The central question is now:

> Is time translation still genuinely gauge after clocks, interactions, quantization or boundaries are added—and what physical states remain when it is not?

That is narrower than “Does conformal gravity solve quantum gravity?” But it is also better posed: each background, charge sector and lifecycle layer can be calculated, audited and potentially falsified.